

A Guideline-Based Symptom and Laboratory Navigator for Cancer Referral Thresholds

Alert burden, cancer specificity and calibration in 37,915 US adults, with the bars set in advance and the failures reported

Rahul (author; surname to be added) · Clinical mentor: Dr. Rishikesh Chavan, UCI Health / CHOC (role and authorship to be agreed) · Oncovision project · Draft of 5 October 2026

Draft for mentor review. Not peer reviewed. The navigator applies the 2015 edition of UK guidance, its rules have **not** been reviewed by a clinician, and nothing here is tested on patients or is medical advice. The key findings are negative: three of four pre-registered bars were missed.

Abstract

Background. Cancers are often found late because early symptoms and mildly abnormal blood tests are individually vague. Clinical guidelines such as the UK's NICE NG12 list symptom-and-test combinations that justify a prompt check, but they are written for clinicians and are hard for a person with a lab report in hand to apply. We built a patient-facing navigator that applies those rules, and measured what it does on real people before anyone relies on it.

Methods. The navigator is a rule engine, not a trained model: 34 rules transcribed from the 2015 edition of NG12, each with its source page, evaluated with three-valued logic so that missing facts are requested rather than assumed. We ran the engine on 37,915 adults aged 40 and over from NHANES III (1988-1994) and ten continuous NHANES cycles (1999-2018), linked to the National Death Index through 2019. Six questions and their pass-or-fail bars were committed to the repository before the analysis. Outcomes were burden of alerts, cancer-specificity among decedents, added value over age and sex in an unseen era, validity of the tool's own iron-deficiency assumption against measured ferritin, and calibration of an age, sex and smoking context layer. Reading level was measured for every sentence the tool can show.

Results. Lab results alone alerted 12% to 17% of people aged 70-79 and 16% to 27% of those 80 and over, almost entirely through the guideline's anaemia-to-stool-test rule, and no more than 2.7% in any age band were told to see a doctor soon. **Three of the four bars with a pass-or-fail criterion were missed.** Among people who died, alerted people were not more likely to have died of cancer (adjusted odds ratio 0.74 (0.63 to 0.88) in 1999-2014; 1.18 (0.91 to 1.54) in NHANES III). Adding the alert to age and sex did not improve prediction of five-year cancer death in the held-out era (AUC gain +0.005, 95% CI -0.009 to +0.019; age and sex alone gave AUC 0.73). The tool's assumption that small red cells indicate iron deficiency had sensitivity 65% and positive predictive value 71% against measured ferritin, and overstated "see a doctor soon" alerts by 47% in people aged 60 and over. The context layer was well calibrated in a later era (slope 0.92). Patient-facing text averaged US grade 6.9.

Conclusions. The navigator applies a published guideline faithfully and readably, but its lab-driven alerts do not carry information about who will die of cancer beyond age and sex. It should therefore be presented as a way of reading a guideline, not as a risk predictor, and lab-only alerts should be worded as questions for a doctor. The

tool is not clinician-reviewed and the rules are UK guidance from 2015; clinician validation is the necessary next step.

Keywords: cancer referral guidelines; symptom checker; laboratory medicine; early detection; NHANES; pre-registration; health literacy.

1. Introduction

A person who has had routine bloodwork usually learns one thing from it: which numbers fall outside a printed range. A blood count and a metabolic panel together produce about thirty values, and what matters in early cancer detection is rarely one of them being wrong. It is a small abnormality in a blood count, alongside a symptom the person has not thought to mention, alongside their age and smoking history. Doctors weigh these together. The person holding the report generally cannot.

Two bodies of work address parts of this. Clinical guidelines, most prominently the NICE guideline NG12 on suspected cancer, list the combinations of age, symptoms and test results that justify an urgent check, chosen so that the chance of cancer for someone meeting a criterion is about three percent or more.¹ Statistical risk models such as QCancer combine symptoms, risk factors and blood tests into a probability.² Both are written for clinicians. Neither is designed to be handed to a person who has a lab report and a worry and wants to know whether it is worth asking about.

Multi-cancer detection from blood alone, in people with no symptoms, is a harder and still unresolved problem. Earlier work in this project built and tested several blood-only panels and withdrew most of them when they failed bars set in advance; the lesson was that routine blood values add little to age and sex for cancer outcomes in the general population. The route that remains well supported is narrower: people who already have a reason to be looking, such as a symptom, where guidelines already define who deserves a prompt check.

This paper describes a navigator for that case and, more importantly, measures it. The questions are practical ones. How many people would the lab rules alert if used without symptoms? Are the alerts specific to cancer, or do they mark ill health in general? Do they add to what age and sex already tell us? Is the tool's own simplifying assumption sound? Can an ordinary reader follow what it says? We committed the questions and the pass-or-fail criteria to the repository before looking at results. Most of them failed, and we report that as the main finding.

2. Methods

2.1 Design

This is a development and retrospective evaluation study. No participant was contacted. All data are public, de-identified US survey records; the study is therefore not human-subjects research under the usual definitions, though the institution's own determination is being sought before any comprehension testing with readers (Section 4.5).

2.2 The rule engine

The navigator (`backend/navigator.py`) reads a request containing age, sex, smoking and asbestos history, a list of symptoms the person has ticked (with how often, where a rule needs it), lab values, and optional findings such as a positive stool test. It evaluates 34 rules held in a plain data file (`backend/navigator_rules.json`) that a clinician can read without reading code. The rules were transcribed from the full NICE NG12 guideline (2015 edition), and every rule records its source page. The engine contains no clinical wording of its own: the action text and the criteria shown to the user come from the file.

Three-valued logic. Every condition evaluates to true, false or unknown. A rule that could apply if one more fact were known is reported as "could apply if", naming exactly what is needed, and is not counted as a match. A missing smoking history or an unentered haemoglobin is therefore asked for and never assumed. Impossible inputs (such as a haemoglobin of 40 g/dL) are refused.

Alert levels. The guideline gives each recommendation a timeframe and a strength. These map to three levels: *see a doctor soon* (urgent or two-week criteria that are recommended), *worth raising* and *mention at a routine visit*. For a given cancer site only the highest level that applies is shown. The tool never states that a person has cancer, and a result of "no threshold met" is worded as different from "nothing is wrong".

How the navigator works

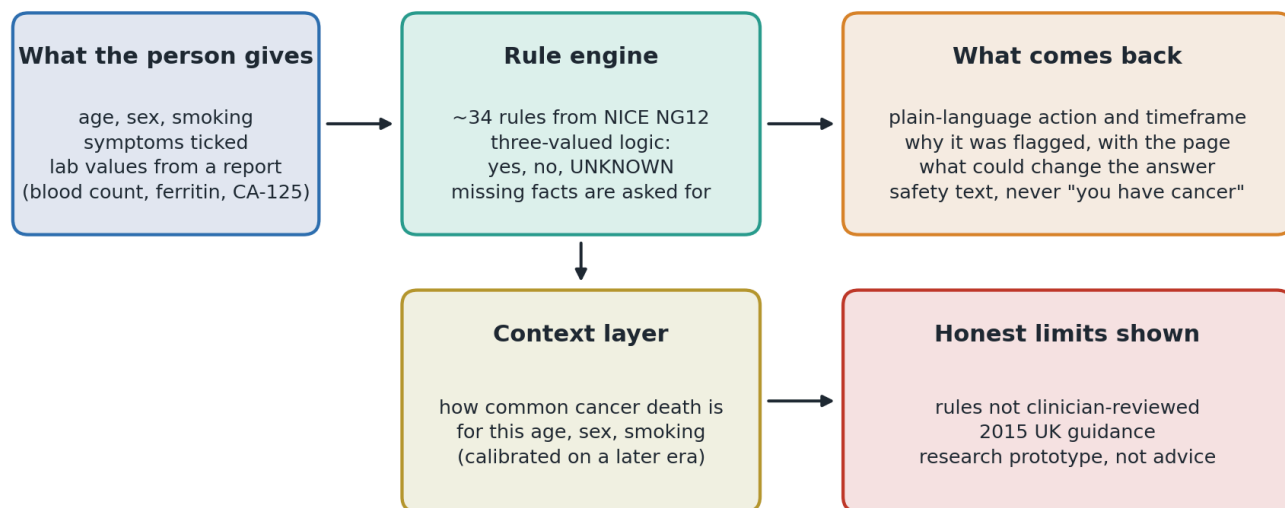


Figure 1. How the navigator works. Inputs on the left are optional; the rule engine asks for what it needs. The context layer and the limits are shown with every answer.

Table 1. The rules, by cancer site.

Cancer site	Rules	Alert levels it can produce
Anal cancer	1	worth raising
Bladder cancer	2	talk soon, mention
Bowel cancer	4	talk soon, worth raising
Breast cancer	3	talk soon, worth raising, mention
Kidney cancer	1	talk soon
Leukaemia	3	talk soon, worth raising
Lung cancer	3	talk soon, worth raising
Lymphoma	2	worth raising
Mesothelioma	1	talk soon
Oesophageal or stomach cancer	3	talk soon, mention
Ovarian cancer	4	worth raising
Pancreatic cancer	2	talk soon, worth raising
Prostate cancer	2	talk soon, mention
Several cancers	2	talk soon
Melanoma	1	talk soon (7-point checklist)

34 rules in total. Each carries its source page in the guideline. Levels come from the guideline's own timeframe and wording ('recommended' versus 'consider').

2.3 Laboratory findings

Raw lab values are converted into the findings the rules read (Table 2). Iron deficiency uses ferritin when it is entered. When it is not, the engine assumes iron deficiency from small red cells (MCV below 80 fL), and says so. Question 4 below tests how reasonable that assumption is.

Table 2. How lab values become the findings the rules read.

Derived finding	Definition used	Where it comes from
Anaemia	Haemoglobin below 13 g/dL (male) or 12 g/dL (female)	WHO definition for adults
Iron deficiency	Ferritin below 15 ng/mL. If no ferritin, MCV below 80 fL is used instead and the person is told so	Common laboratory cut-off; MCV rule is the tool's disclosed assumption
Iron-deficiency anaemia	Anaemia and iron deficiency together	Derived
Thrombocytosis	Platelets above $400 \times 10^9/L$	Guideline
Raised white count	White cells above $11 \times 10^9/L$	Common laboratory cut-off
Raised CA-125	35 U/mL or more	Guideline

2.4 Data

NHANES is a continuing US national survey with a physical examination and laboratory tests. NCHS links participants to the National Death Index, giving vital status and underlying cause of death through 31 December 2019. We used NHANES III and ten continuous cycles; cycles examined in 2015–2018 have under five years of follow-up for many people and so were used only for questions that need no outcome (Table 3). Participants who reported a previous cancer diagnosis were excluded where the questionnaire asked, and nobody was excluded for how they later died, a choice learned from an earlier cohort in this project that dropped everyone who died of another cause and thereby made every early death a cancer death. The outcome is **death from cancer** (NCHS underlying cause 2) within 60 months of the examination: a later and harsher endpoint than diagnosis, and one that cannot show whether an alert would have led to a correct referral.

Table 3. The data.

Source	Years	Adults 40+ with a blood count	Used for
NHANES III, linked to the National Death Index	1988–1994	9,335	Burden; cancer specificity; added value (external era)
Continuous NHANES, eight cycles with five-year follow-up	1999–2014	22,583	All questions; context layer
Continuous NHANES, two newest cycles	2015–2018	5,997	Burden and ferritin check only (follow-up too short)
Total	1988–2018	37,915	

NHANES: US National Health and Nutrition Examination Survey. Mortality follow-up through 31 December 2019 from the NCHS public-use linked mortality files.

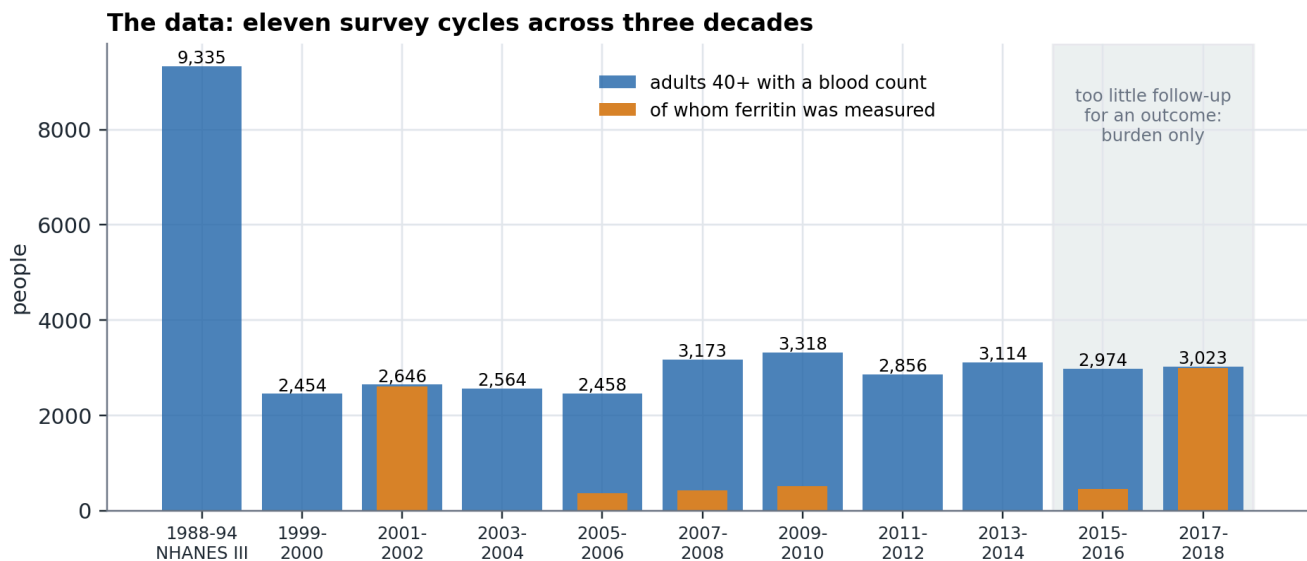


Figure 2. The data. Blue bars are adults aged 40 or over with a blood count. Orange shows how many also had ferritin measured, which differs by survey cycle. The shaded cycles have too little follow-up for an outcome.

2.5 What the engine was given

The real engine was run, not a re-implementation. Two versions were scored for every person. **Labs only** passed haemoglobin, MCV, platelets, white count and (where measured) ferritin, with no symptoms and no smoking history. **Labs plus proxies** added smoking (100 or more cigarettes in a lifetime) and three symptom stand-ins, passed only when present: *weight loss* (weight a year ago minus weight now of at least 5% of the earlier weight and not trying to lose weight, or at least 10% where that question was not asked), *chronic cough* (cough on most days for three months, asked 1999-2012) and *breathlessness* (asked of people 40 and over). NHANES never asked about bleeding, lumps, bowel change, appetite, tiredness or jaundice, so most of the guideline's symptoms are invisible here, and every alert rate below is a **lower bound** on what someone reporting those symptoms would see. The proxies are crude: weight loss in a survey includes illness and medication effects that a person ticking "no known reason" would exclude.

2.6 Pre-registered questions and bars

Six questions and their criteria were committed before analysis (docs/NAVIGATOR_EVIDENCE_PREREG.md) and are summarised with the results in Table 4. The earlier project had repeatedly found that results look better when the comparison is weak, so every prediction question uses age and sex as the baseline, and cancer specificity is tested only among people who died, so that "sick people die of things" cannot masquerade as a cancer signal.

2.7 Statistics

Proportions use Wilson 95% intervals.³ Cancer specificity among decedents (deaths within ten years of examination) used logistic regression of cancer versus other cause on the flag, age, age squared, sex and examination year, with Wald intervals. Added value used logistic models of five-year cancer death with age, age squared and sex, with and without the alert, fitted on 1999-2006 and tested on 2007-2014 (and, exploratorily, fitted on 1999-2014 and tested on NHANES III); the AUC difference was bootstrapped 1,000 times, resampling cases and non-cases separately. Non-cancer deaths inside the window are counted as non-cases, which is a competing-risk simplification. All analyses are unweighted, because the aim was to exercise the engine on realistic profiles, not to estimate a national prevalence.

2.8 Context layer

To answer "how much of this is just my age", the tool reports how common five-year cancer death is among US adults of the same age, sex and smoking history: a logistic model of age, age squared, sex and smoking, fitted on 1999-2006 and tested on 2007-2014 (bar: calibration slope between 0.8 and 1.2), then refitted on all outcome cycles for use in the tool. It does not use labs or symptoms and is shown as a group average, not the person's risk.

2.9 Readability

Every sentence the tool can show was scored with the Flesch-Kincaid grade level.⁴ The bar, set before the first measurement, was a mean of at most 8 and no action line above 10. This is a floor and not a test of understanding.

2.10 Software and reproducibility

Backend: Python and FastAPI; frontend: Next.js, deployed publicly. There are 79 automated tests, including properties such as "a symptom-free person with normal labs is never told to see a doctor soon over 300 random profiles", "the tool never says 'you have cancer'" and "a rule that refers to a symptom the form cannot ask about is a loud failure". Every number in this paper is regenerated by scripts in the repository, and the tables and figures are built directly from their output. Reporting follows the spirit of TRIPOD+AI where it applies;⁵ because the engine is not a trained model there is no training-set leakage to guard against in the rules themselves, only in the context layer, which is validated out of era.

3. Results

3.1 Summary against the pre-registered bars

Table 4. Pre-registered questions and what happened.

#	Pre-registered question and bar	Result	Outcome
1	Burden is stable across eras: 'see a doctor soon' at most 3% and any alert at most 20% in every age band and era	Worst 'soon' 2.7%; worst any-alert 26.7% (age 80+, 2015-2018)	Missed (any-alert, ages 80+)
2	Alerted people are more likely to die of cancer than of something else, among decedents (adjusted OR interval above 1 in 1999-2014, and OR above 1 in NHANES III)	1999-2014: OR 0.74 (0.63 to 0.88). NHANES III: OR 1.18 (0.91 to 1.54)	Missed
3	An alert adds to age and sex for five-year cancer death in the held-out era (AUC-gain interval excludes 0)	Labs and proxy symptoms: +0.005 (-0.009 to +0.019); labs only +0.000	Missed
4	Report how well MCV below 80 stands in for ferritin (no bar)	Sensitivity 65%, PPV 71%	Reported
5	Which rules fire (no bar)	Colorectal rules account for almost all lab-driven alerts	Reported
6	Age, sex and smoking context layer is calibrated in a later era (slope 0.8 to 1.2)	Slope 0.92; AUC 0.747	Met

Bars were committed to the repository before the analysis was run (docs/NAVIGATOR_EVIDENCE_PREREG.md).

3.2 Alert burden

With lab results alone and no symptoms, the share of adults alerted rises steeply with age (Table 5, Figure 3): about 4% to 8% at 40-49, 1% to 2% at 50-59, and 16% to 27% at 80 and over. The pattern is close to identical in all four eras, spanning thirty years of laboratory methods. "See a doctor soon" stayed at or below 2.7% in every band and era, and is zero below age 60.

Almost every lab-driven alert comes from one guideline rule: anaemia in people 60 or over, or iron-deficiency anaemia under 60, leads to a stool test. In the two newest cycles, where ferritin was measured for most people, this rule alerted 13.7% of women but 1.6% of men aged 40-49, a difference consistent with menstrual iron loss, which the rule does not take into account. That is a real feature of the guideline applied to people who may have a benign explanation. The pre-registered 20% line for any alert was crossed only at age 80 and over, in three of the four eras (21%, 23% and 27%).

How many symptom-free adults the lab rules alert, by age and era

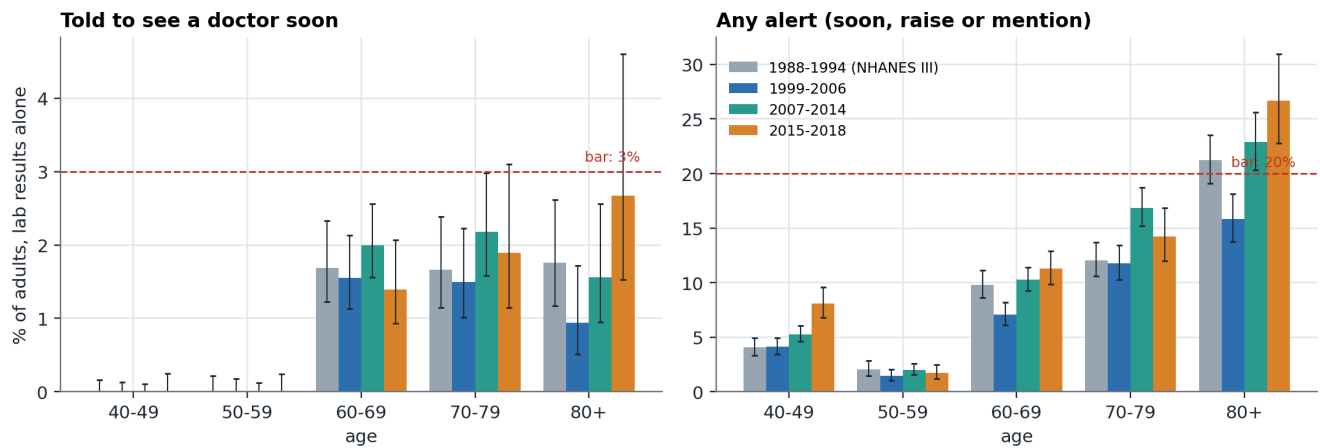


Figure 3. Share of symptom-free adults alerted by lab results alone, by age and era. Red dashed lines are the pre-registered bars. Error bars are 95% Wilson intervals.

Table 5. Share of adults alerted by lab results alone, by era and age.

Era	Age	Adults	See a doctor soon	Any alert (95% interval)
1988-1994 (NHANES III)	40-49	2,414	0.0%	4.1% (3.3 to 4.9)
	50-59	1,746	0.0%	2.1% (1.5 to 2.8)
	60-69	2,183	1.7%	9.8% (8.6 to 11.1)
	70-79	1,683	1.7%	12.1% (10.6 to 13.7)
	80+	1,309	1.8%	21.2% (19.1 to 23.5)
1999-2006	40-49	2,932	0.0%	4.1% (3.5 to 4.9)
	50-59	2,138	0.0%	1.5% (1.0 to 2.1)
	60-69	2,388	1.6%	7.1% (6.1 to 8.2)
	70-79	1,597	1.5%	11.8% (10.3 to 13.5)
	80+	1,067	0.9%	15.8% (13.8 to 18.1)
2007-2014	40-49	3,565	0.0%	5.3% (4.6 to 6.1)
	50-59	3,187	0.0%	2.0% (1.6 to 2.6)
	60-69	3,046	2.0%	10.3% (9.2 to 11.4)
	70-79	1,701	2.2%	16.9% (15.2 to 18.7)
	80+	962	1.6%	22.9% (20.3 to 25.6)
2015-2018	40-49	1,522	0.0%	8.1% (6.8 to 9.6)
	50-59	1,575	0.0%	1.7% (1.2 to 2.5)
	60-69	1,657	1.4%	11.3% (9.8 to 12.9)
	70-79	793	1.9%	14.2% (12.0 to 16.9)
	80+	450	2.7%	26.7% (22.8 to 30.9)

Lab results alone, no symptoms entered. Intervals are Wilson 95% intervals.

3.3 Are the alerts specific to cancer?

No. Among the 3,864 people in 1999-2014 who died within ten years, those alerted by labs alone were *less* likely to have died of cancer than of something else (13.5% versus 21.7%; adjusted odds ratio 0.63 (0.49 to 0.80)). Adding the proxy symptoms moved the estimate to 0.74 (0.63 to 0.88). The result did not replicate in direction in NHANES III (1.18 (0.91 to 1.54), interval including 1). In no era did an alert identify cancer deaths.

Looking at one flag at a time (Table 7, Figure 4; exploratory), anaemia was the clearest: among decedents, anaemic people had about 40% lower odds of having died of cancer rather than another cause (OR 0.57 (0.45 to 0.72)). The proxy for weight loss and breathlessness pointed the same way. Smoking, the one flag that is not a symptom or lab, pointed the expected way (OR 1.30 (1.09 to 1.56)).

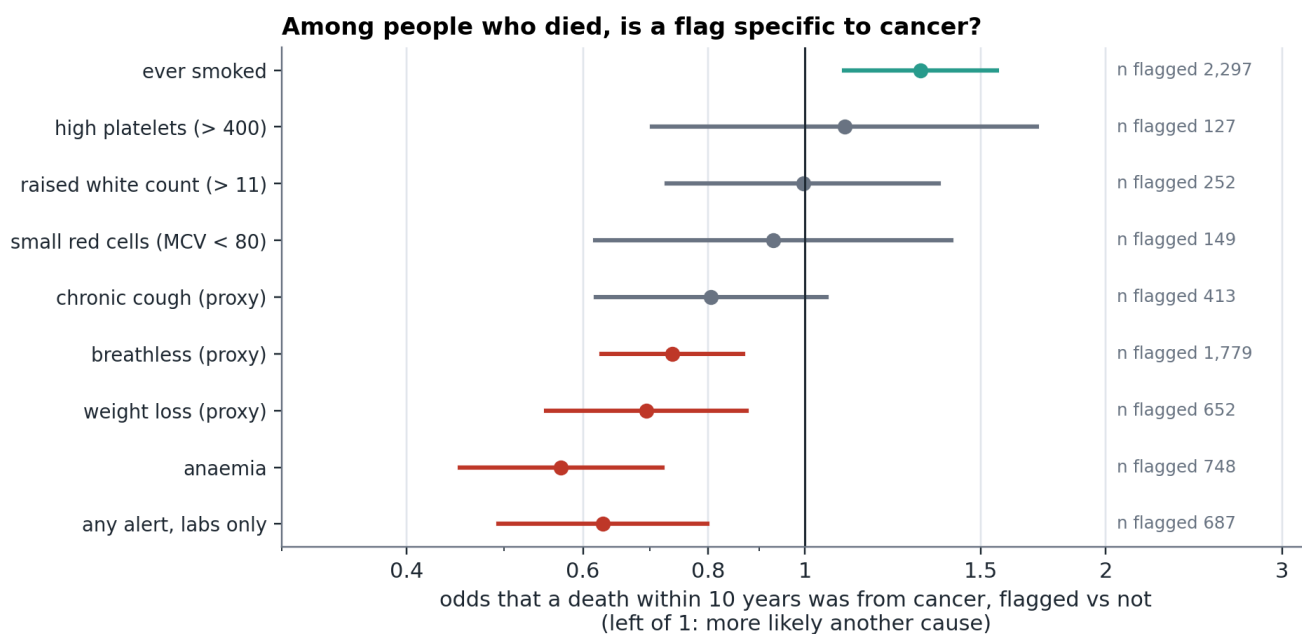


Figure 4. Among people who died within ten years, the odds that the death was from cancer rather than another cause, for people with each flag versus those without. Red intervals lie entirely below 1. Adjusted for age, sex and year.

Table 6. Cancer specificity among people who died within ten years.

Group	Deaths in 10 years	Alerted	Cancer share, alerted	Cancer share, not alerted	Adjusted OR (95% CI)
1999-2014 labs only	3,864	687	13.5%	21.7%	0.63 (0.49 to 0.80)
1999-2014 labs and proxies	3,864	2,094	18.5%	22.3%	0.74 (0.63 to 0.88)
NHANES III labs only	2,324	400	23.0%	22.4%	1.18 (0.91 to 1.54)

Decedents only: everyone here died within ten years, so the question is whether the death was from cancer. Adjusted for age, age squared, sex and (pooled) examination year.

Table 7. Each flag on its own, among decedents (exploratory).

Single flag	Decedents flagged	Adjusted OR for dying of cancer rather than another cause (95% CI)
anaemia	748 of 3,864	0.57 (0.45 to 0.72)
small red cells (MCV < 80)	149 of 3,864	0.93 (0.62 to 1.40)
high platelets (> 400)	127 of 3,864	1.10 (0.70 to 1.71)
raised white count (> 11)	252 of 3,864	1.00 (0.73 to 1.36)
weight loss (proxy)	652 of 3,864	0.69 (0.55 to 0.87)
chronic cough (proxy)	413 of 3,864	0.81 (0.62 to 1.05)
breathless (proxy)	1,779 of 3,864	0.74 (0.63 to 0.87)
ever smoked	2,297 of 3,864	1.30 (1.09 to 1.56)
any alert, labs only	687 of 3,864	0.63 (0.49 to 0.80)

Exploratory. Nine comparisons; only intervals well away from 1 should be believed. 'Proxy' means a crude NHANES stand-in for a symptom (see Methods).

3.4 Do alerts add to age and sex?

Age and sex alone predicted five-year cancer death with AUC 0.73 in the held-out era. Adding the labs-only alert changed this by +0.000 (95% CI -0.005 to +0.005); adding the alert from labs and proxy symptoms changed it by +0.005 (-0.009 to +0.019). Both intervals include zero, so the bar was missed. In the exploratory NHANES III test the gain was +0.009 (+0.005 to +0.014), statistically above zero but under one point of AUC and not in the pre-registered test.

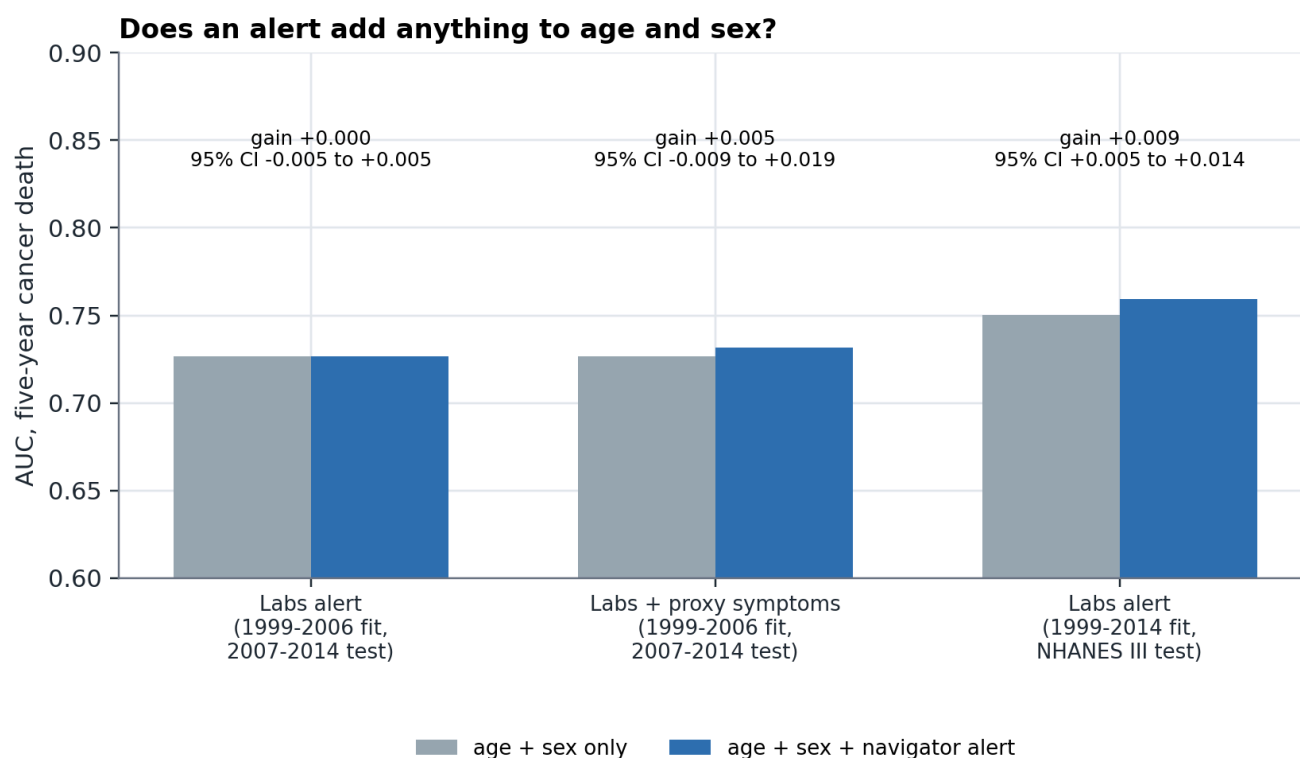


Figure 5. AUC for five-year cancer death with and without the navigator alert. Age and sex do nearly all the work.

Table 8. Does the alert add to age and sex for five-year cancer death?

Comparison	Test people	Cancer deaths	AUC, age + sex	AUC, with alert	Gain (95% CI)
Labs alert; fit 1999-2006, test 2007-2014	12,438	179	0.727	0.727	+0.000 (-0.005 to +0.005)
Labs + proxy symptoms; fit 1999-2006, test 2007-2014	12,438	179	0.727	0.732	+0.005 (-0.009 to +0.019)
Labs alert; fit 1999-2014, test NHANES III (exploratory)	9,326	238	0.750	0.759	+0.009 (+0.005 to +0.014)

AUC: probability that a person who died of cancer within five years has a higher score than one who did not. Bootstrap intervals, 1,000 resamples.

3.5 The tool's own iron-deficiency assumption

Among 7,373 adults with a measured ferritin, 817 were anaemic and 328 of those were iron-deficient by ferritin. Using MCV below 80 as a stand-in caught 65% of them and was right 71% of the time. Among 2,743 adults 60 or over with a ferritin, the proxy told 53 to see a doctor soon where measured ferritin told 36. A person who enters ferritin gets a more accurate answer than one who does not, and the tool says which one they got.

Testing the tool's own disclosed assumption

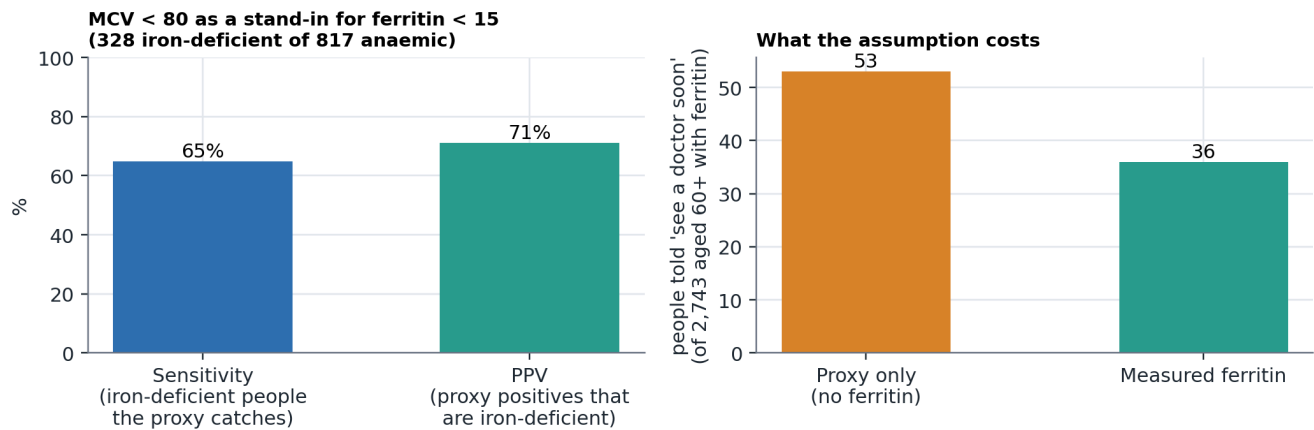


Figure 6. Left: how well MCV below 80 stands in for ferritin below 15 among anaemic adults. Right: the cost in alerts for people aged 60 and over.

3.6 Which rules fire

Table 9. Rules that fired, labs plus proxy symptoms.

Rule	People it fired for	Died of cancer within 5 years	Mean age
Chest X-ray (lung)	5,953	2.3%	60
Stool test for hidden blood (bowel)	3,175	2.6%	66
Unexplained weight loss, several cancers	2,520	1.8%	60
Consider ovarian tests	1,446	1.2%	60
Urgent bowel referral	184	3.3%	70
Consider chest X-ray (lung)	127	0.8%	58
Consider endoscopy (stomach, gullet)	55	3.6%	69
Everyone in the analysis	22,560	1.5%	

Labs plus proxy symptoms, 1999-2014. Descriptive only: the groups differ in age, and a higher rate in a group is mostly its older age.

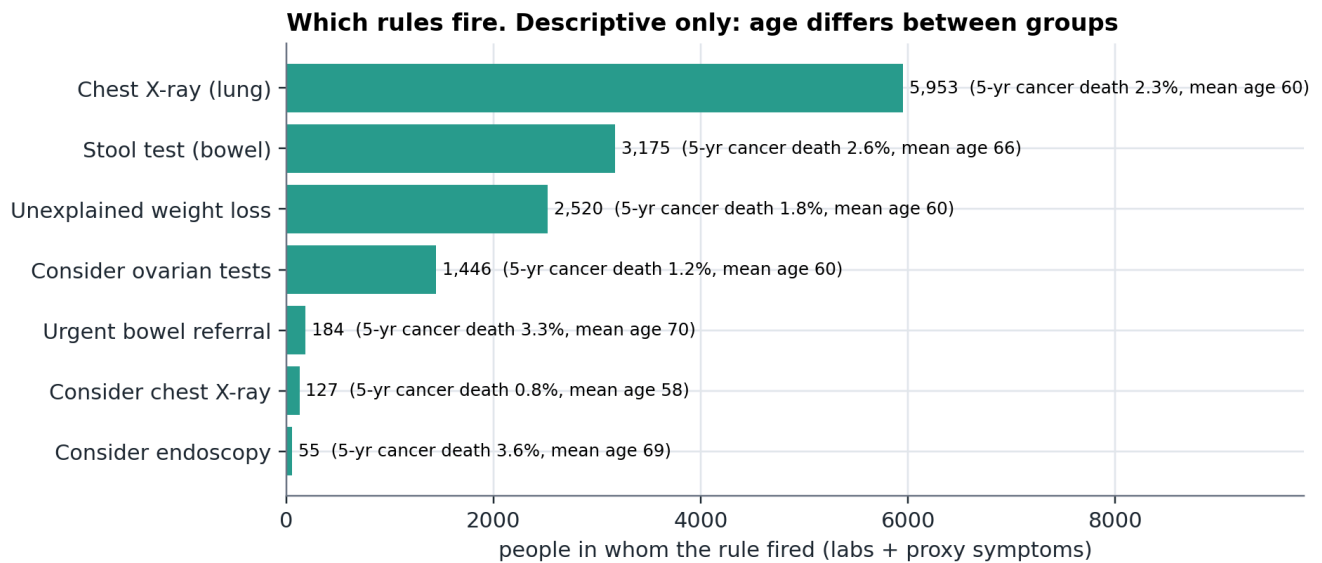


Figure 7. Which rules fire for labs plus proxy symptoms. The lung rule fires most often because breathlessness and smoking are common in older NHANES adults, not because of an excess of cancer.

3.7 The age context layer

Fitted on 1999-2006 and tested on 2007-2014, the age, sex and smoking model had AUC 0.747 and a calibration slope of 0.92, inside the pre-registered 0.8 to 1.2. Its Brier score (0.01398) was slightly better than predicting the average for everyone (0.01419). The model was refitted on 24,180 adults with 379 five-year cancer deaths for use in the tool. Typical outputs range from about 2 in 1,000 for a 45-year-old woman who has never smoked to about 50 in 1,000 for a 75-year-old man who has smoked.

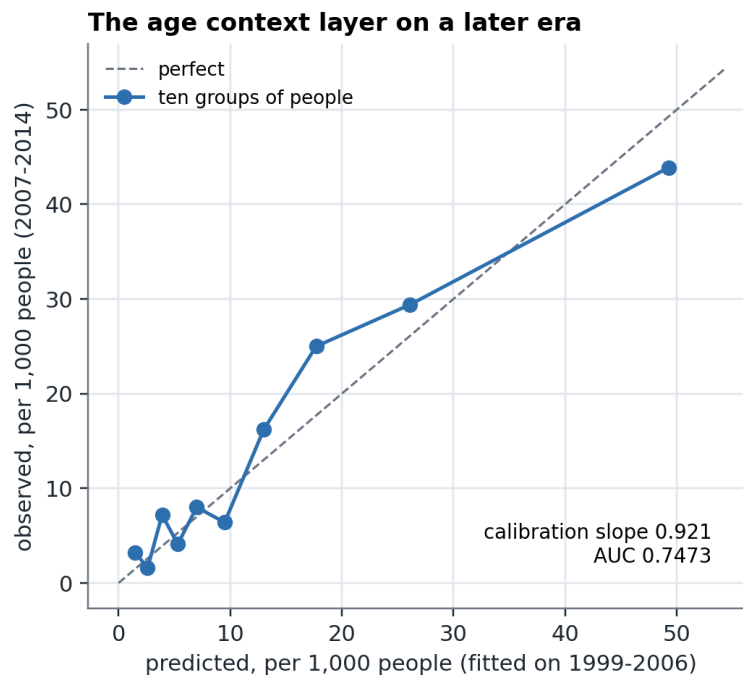


Figure 8. Calibration of the context layer in the later era.

Table 10. Calibration of the context layer by tenth.

Predicted per 1,000	Observed per 1,000	People
1.5	3.2	1,250
2.6	1.6	1,268
3.9	7.2	1,254
5.3	4.1	1,208
7.0	8.0	1,255
9.5	6.4	1,249
13.0	16.2	1,298
17.7	25.0	1,202
26.1	29.4	1,225
49.3	43.9	1,229

Ten equal-sized groups of the 2007-2014 people, ordered by predicted risk, from a model fitted only on 1999-2006.

3.8 Readability

Patient-facing action and headline lines averaged US grade 6.9, with none over grade 10. The first measurement missed the bar: five action lines exceeded grade 10 and the emergency advice scored grade 16.5 because it was a single long sentence. After rewriting without changing meaning, the emergency advice scored grade 4.8. The remaining hard items in the table are one-word symptom labels, where the formula over-scores.

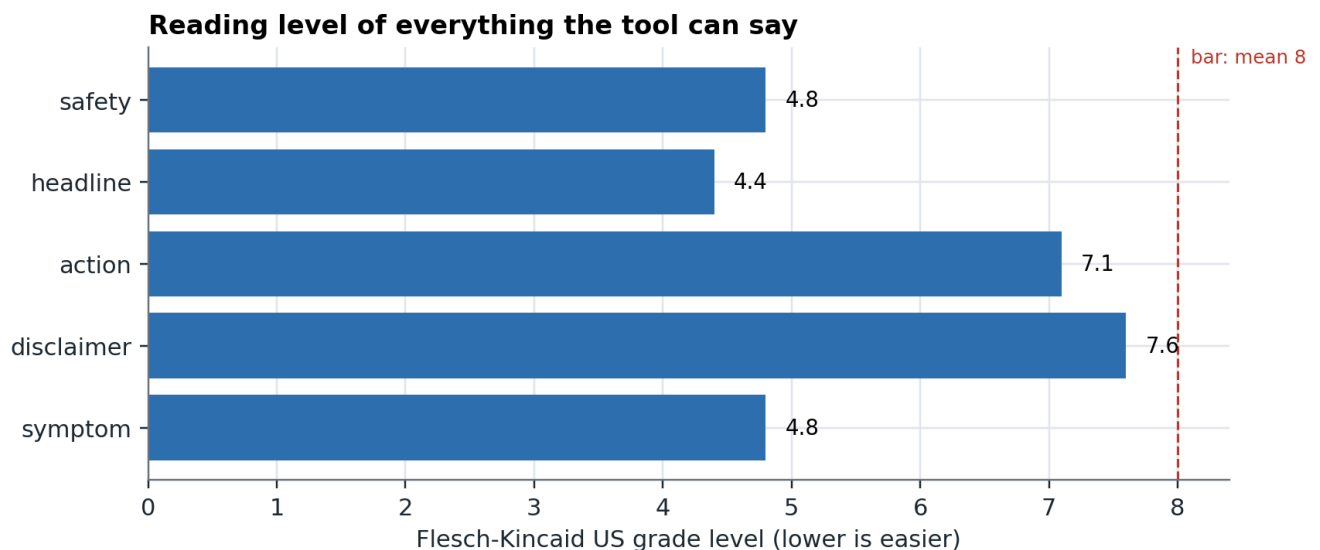


Figure 9. Reading level of each kind of text. Lower is easier; the dashed line is the bar.

Table 11. Reading level of the tool's text.

Text the tool can show	Items	Mean US grade	Hardest item	Over grade 10
Action	34	7.1	9.1	0
Headline	2	4.4	5.8	0
Safety	2	4.8	5.3	0
Disclaimer	1	7.6	7.6	0
Symptom	53	4.8	32.0	5

Flesch-Kincaid grade level. 'Symptom' items are one-line labels, where the formula over-scores very short sentences; five of them exceed grade 10 for that reason only.

3.9 The tool as a person sees it

Figures 10 and 11 are screenshots of the deployed application. The case entered is invented for illustration: a 66-year-old man who has smoked, with unexplained weight loss and a blood count showing iron-deficiency anaemia.

← Back to Oncovision

Symptom and lab pattern navigator

Tell it what you have noticed and, if you have them, your lab results. It checks whether the combination meets a published guideline threshold for a prompt check, and says what to ask for.

⚠ RESEARCH PROTOTYPE. NOT MEDICAL ADVICE.

It applies UK guideline thresholds (the 2015 edition) that a clinician has not yet reviewed for this tool, and it cannot examine you. It never says you have cancer, and meeting no threshold does not mean nothing is wrong. If something worries you, or is getting worse, see a doctor.

Coughing or vomiting a lot of blood, severe chest pain, trouble breathing, or black bloody stools with faintness need urgent care now, not this page.

About you

AGE: years | SEX: Not given | EVER SMOKED: Not given | ASBESTOS EXPOSURE: Not given

What you have noticed

Tick only things with no known cause. If something has an explanation (an injury, a diagnosed condition, a medicine), do not tick it.

CHEST AND BREATHING

- ☐ Coughing up blood
Blood in the phlegm you cough up.
- ☐ A cough that has no known cause
- ☐ Shortness of breath with no known cause
- ☐ Chest pain with no known cause
- ☐ Chest infections that keep coming back or will not clear

Figure 10. The entry form, with the research-prototype warning at the top.

WORTH SEEING A DOCTOR SOON

This combination meets a guideline threshold for a prompt check. Most people who meet it do not have cancer, but it is worth seeing a doctor soon.

Bowel Cancer **SEE A DOCTOR SOON**

Ask your doctor for a specialist appointment within about 2 weeks to check for bowel cancer.

WHY THIS WAS FLAGGED

- aged 60 or over
- iron-deficiency anaemia

Source: NG12 2015, colorectal cancer, p.137

Lung Cancer **SEE A DOCTOR SOON**

Ask for a chest X-ray within about 2 weeks to check for lung cancer.

WHY THIS WAS FLAGGED

- aged 40 or over
- have ever smoked
- any one of: a cough that has no known cause; shortness of breath with no known cause; chest pain with no known cause; weight loss with no known reason; loss of appetite with no known reason; tiredness or fatigue with no known cause

Source: NG12 2015, lung and pleural cancers, p.45

Several Cancers **SEE A DOCTOR SOON**

Weight loss with no known reason can be a sign of several cancers. These include bowel, stomach, gullet, lung, prostate, pancreas, kidney and bladder cancer. It can also have other causes. Guidelines say a doctor should look for the cause soon.

WHY THIS WAS FLAGGED

- weight loss with no known reason

Source: NG12 2015, symptoms that are not specific to one cancer, p.347

Figure 11. An answer for the invented case: what was flagged, why, and where in the guideline it comes from, followed by the age context.

4. Discussion

4.1 What the results mean

The navigator did what it was designed to do. It applied a published guideline to a person's facts, asked for what was missing, explained each alert with the page it came from, and did so at a reading level most adults can follow. It did not do something it was never designed to do and that a patient-facing tool might be mistaken for: it did not identify who would die of cancer. This distinction is the main lesson. The NG12 criteria are thresholds for *looking*, set so that the chance of finding cancer is about three percent or more among people who have *already come to a doctor with a symptom*. Applied to symptom-free people using only their lab values, the same thresholds mostly flag the common conditions that cause anaemia and raised white counts.

The decedent-only result needs careful reading. It does not show that anaemia protects against cancer death, and it does not show the guideline is wrong. Anaemia is common in people dying of kidney, heart and lung disease, and in a general-population sample those causes of death are far more common than cancer. Whether an alerted person who then received a stool test or a colonoscopy would have had a bowel cancer found is not something death records can show. What the result does show is that a tool which presents lab-only alerts as a signal of cancer would mislead.

The context layer is the other practical result. Age and sex alone give an AUC of about 0.73 for five-year cancer death, and nothing the navigator adds changes that appreciably. Showing a person what is ordinary for their age is cheap, calibrated and, we believe, useful for reading an alert in proportion; whether it helps or alarms people is a question for readers, not for this data.

4.2 Changes made because of the results

Lab-only alerts now carry a note that common causes are not cancer and that, in a US survey, these findings did not help tell who later died of cancer once age and sex were known. A lower-tier alert for a cancer site is no longer shown beside a higher-tier one. The context layer was added to the tool because its bar was met. A rule defect found along the way (asbestos wording shown to a smoker without exposure) was fixed and recorded. These changes are listed as deviations in the pre-registration, which was committed before the first analysis.

4.3 Relation to existing work

QCancer-style models report AUCs of roughly 0.84 to 0.88 for cancer when symptoms, risk factors and blood tests are combined in primary-care records.² Those models use a much richer symptom set and diagnoses as the outcome; the figures are not comparable with an AUC for cancer death in a survey, and we make no claim to match them. The navigator's different contribution is access: a person can enter a lab report and symptoms, see why an alert arose in the guideline's own words, and read a plain-language answer. We have not verified that no similar patient-facing tool exists, and the novelty claim should be checked before it is made.

4.4 Limitations

- **Not clinician-reviewed.** The rule file records `review_status: not yet reviewed`, and the tool shows this to every user. Transcription errors are possible. A design decision that departs from the guideline text (gating the lung "consider chest X-ray" thrombocytosis path behind at least one symptom, because applied literally it alerted 3.7% of symptom-free adults over 40) needs a clinician's confirmation.
- **Dated and foreign guidance.** The rules are the 2015 edition of UK guidance, which has since been revised (for example, the bowel pathway now uses the faecal immunochemical test). US guidance may differ.
- **Death, not diagnosis.** The outcome cannot show whether an alert would have led to a correct referral. Cancers that were found early and cured are counted as non-cases.
- **Proxies for symptoms.** NHANES has three crude symptom stand-ins and none of the symptoms that drive most of the guideline's high-value rules (bleeding, lumps, bowel change, jaundice). Results for symptom-plus-lab combinations are therefore weak evidence.
- **Self-reported weight and unweighted analysis.** The proxy for weight loss misclassifies people. The analyses ignore survey weights and describe the sample, not the US population.
- **Competing risks.** Non-cancer deaths inside the window are treated as non-cases, which a cause-specific hazard model would handle more carefully.
- **Single comparison for the main prediction question.** The one analysis on NHANES III went the other way from the pre-registered era, and with 238 cancer deaths and a one-point AUC difference it should not be over-read.
- **Reading level is not comprehension.** A short sentence can still be misunderstood.

4.5 Next steps

1. Clinician sign-off of every rule against the current guideline, recorded in the rule file.
2. A vignette study in which clinicians independently apply the guideline text to written cases and their answers are compared with the tool's, with under-alerting counted separately because it is the harmful direction (docs/NAVIGATOR_VALIDATION_PROTOCOL.md).
3. A comprehension check with lay readers, after an ethics determination.
4. An evaluation on data with real symptoms and diagnoses, such as a clinical records database, once access is obtained.
5. Reconciling the rules with current UK and US guidance.

5. Conclusion

A guideline-based navigator can be built, can be made readable, and can be tested against bars set in advance. In this test it applied the guideline as written and produced a stable, modest alert burden. Its lab-driven alerts did not identify who would die of cancer, three of its four pass-or-fail bars were missed, and the paper says so. The honest claim is therefore a modest one: the navigator is a faithful, readable way to apply a published referral guideline, accompanied by a calibrated statement of how common cancer death

is for a person's age, and it should not be presented as a detector. Whether it helps real people reach a diagnosis sooner is untested.

Declarations

AI assistance

This project was developed with the assistance of an AI model (Anthropic Claude, Sonnet and earlier versions, via Claude Code). The AI was used to write and test code, to run analyses, to draft text, and to search for background literature. An AI cannot be an author. The human author is responsible for the work, reviewed the results, and must verify every statement and reference before submission; the references listed here were written from the author's and the assistant's knowledge of the literature and have not yet been checked against the originals.

Ethics

The analyses use public, de-identified records and involve no contact with participants. Any study recruiting readers or patients will need an ethics determination first.

Data and code availability

Code, rules, tests, pre-registration and all result files are in the public repository (github.com/palashraksafk/OncoVision). NHANES data are public from the US National Center for Health Statistics; the linked mortality files are public-use. The navigator is deployed at oncovisionai.vercel.app/navigator.

Clinical mentorship and authorship

Clinical mentorship is being sought from Dr. Rishikesh Chavan (UCI Health / CHOC). His role, review of the rules and authorship status are to be agreed with him and are not asserted here.

Conflicts of interest and funding

None declared. No funding.

Warning

The navigator is a research prototype and is not medical advice. A person who is worried about a symptom should see a doctor and should seek urgent care for severe symptoms regardless of what the tool says.

References

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All references are to be checked against the originals before submission.

Appendix: the rules

The full rule file, with every criterion and source page, is `backend/navigator_rules.json`. A reviewing clinician can read it directly. Each rule's action text, as the tool shows it to a person, is listed below with its source page; the criteria behind each are in the file.

Rule	Site	Level	What the tool tells the person	Source
lung_haemoptysis	lung cancer	talk soon	Ask your doctor for a specialist appointment within about 2 weeks to check for lung cancer.	NG12 2015, lung and pleural cancers, p.45
lung_xray_symptoms	lung cancer	talk soon	Ask for a chest X-ray within about 2 weeks to check for lung cancer.	NG12 2015, lung and pleural cancers, p.45
lung_xray_consider	lung cancer	worth raising	Ask your doctor if a chest X-ray within about 2 weeks is worth doing.	NG12 2015, lung and pleural cancers, p.45
meso_xray_symptoms	mesothelioma	talk soon	Ask for a chest X-ray within about 2 weeks. Doctors check for mesothelioma when these symptoms come with asbestos exposure.	NG12 2015, lung and pleural cancers, p.49
upper_gi_urgent	oesophageal or stomach cancer	talk soon	Ask for a camera test of the gullet and stomach (an endoscopy) within about 2 weeks.	NG12 2015, oesophageal and stomach cancers, pp.62 and 84
upper_gi_haematemesis	oesophageal or stomach cancer	mention	Tell a doctor. A non-urgent endoscopy may be considered. Vomiting a lot of blood is an emergency: seek care now.	NG12 2015, oesophageal and stomach cancers, pp.62 and 84
upper_gi_consider	oesophageal or stomach cancer	mention	Ask whether a non-urgent endoscopy is worth doing.	NG12 2015, oesophageal and stomach cancers, pp.62 and 84
pancreas_jaundice	pancreatic cancer	talk soon	Ask your doctor for a specialist appointment within about 2 weeks.	NG12 2015, pancreatic cancer, p.71
pancreas_scan	pancreatic cancer	worth raising	Ask whether an urgent scan (CT, or ultrasound if CT is not available) is worth doing.	NG12 2015, pancreatic cancer, p.71
colorectal_urgent	bowel cancer	talk soon	Ask your doctor for a specialist appointment within about 2 weeks to check for bowel cancer.	NG12 2015, colorectal cancer, p.137
colorectal_mass	bowel cancer	worth raising	Ask your doctor if you need a specialist appointment within about 2 weeks.	NG12 2015, colorectal cancer, p.137
colorectal_young_bleeding	bowel cancer	worth raising	Ask your doctor if you need a specialist appointment within about 2 weeks.	NG12 2015, colorectal cancer, p.137
colorectal_stool_test	bowel cancer	worth raising	Ask for a stool test for hidden blood.	NG12 2015, colorectal cancer, p.137

Rule	Site	Level	What the tool tells the person	Source
anal_mass	anal cancer	worth raising	Ask your doctor if you need a specialist appointment within about 2 weeks.	NG12 2015, anal cancer, p.144
breast_urgent	breast cancer	talk soon	A doctor should arrange an urgent breast clinic appointment (within about 2 weeks).	NG12 2015, breast cancer, p.155
breast_consider	breast cancer	worth raising	Ask whether an urgent breast clinic appointment (within about 2 weeks) is needed.	NG12 2015, breast cancer, p.155
breast_young_lump	breast cancer	mention	Ask for a non-urgent referral to have the lump checked.	NG12 2015, breast cancer, p.155
ovarian_symptoms	ovarian cancer	worth raising	Ask for a CA-125 blood test. These symptoms, especially after 50 and on more than about 12 days a month, are the ones doctors test for ovarian cancer.	NG12 (ovarian cancer recommendations dated 2011), p.159
ovarian_ibs	ovarian cancer	worth raising	Ask for tests for ovarian cancer. Irritable bowel syndrome rarely appears for the first time at this age.	NG12 (ovarian cancer recommendations dated 2011), p.159
ovarian_ca125	ovarian cancer	worth raising	A CA-125 at or above 35 with these symptoms means an ultrasound of the abdomen and pelvis should be arranged.	NG12 (ovarian cancer recommendations dated 2011), p.159
ovarian_other	ovarian cancer	worth raising	Ask whether tests for ovarian cancer are worth doing, alongside looking for other causes.	NG12 (ovarian cancer recommendations dated 2011), p.159
prostate_psa	prostate cancer	talk soon	Ask your doctor for a specialist appointment within about 2 weeks.	NG12 2015, prostate cancer, p.178
prostate_symptoms	prostate cancer	mention	Ask about a PSA blood test and a prostate examination.	NG12 2015, prostate cancer, p.178
bladder_urgent	bladder cancer	talk soon	Ask your doctor for a specialist appointment within about 2 weeks.	NG12 2015, bladder cancer, p.189
bladder_uti	bladder cancer	mention	Ask for a non-urgent referral to look for a cause of repeated urine infections.	NG12 2015, bladder cancer, p.189
renal_haematuria	kidney cancer	talk soon	Ask your doctor for a specialist appointment within about 2 weeks.	NG12 2015, renal cancer, p.203
leukaemia_children_blood_count	leukaemia	talk soon	Ask for a blood count within about 2 days. In children and young people these symptoms are the ones doctors check a blood count for, to look for leukaemia.	NG12 2015, leukaemia, p.256

Rule	Site	Level	What the tool tells the person	Source
leukaemia_children_petechiae	leukaemia	talk soon	Tiny red or purple spots that do not fade under pressure in a child or young person need to be seen by a specialist straight away. Seek urgent care now.	NG12 2015, leukaemia, p.256
leukaemia_adult_blood_count	leukaemia	worth raising	Ask whether a blood count within about 2 days is worth doing. Most people with tiredness or similar symptoms do not have leukaemia.	NG12 2015, leukaemia, p.256
lymphoma_adult	lymphoma	worth raising	Ask your doctor if you need a specialist appointment within about 2 weeks. Fever, night sweats, breathlessness, itching, or weight loss make it more likely.	NG12 2015, non-Hodgkin and Hodgkin lymphoma, pp.274 and 280
lymphoma_children	lymphoma	worth raising	Ask whether a specialist appointment within about 2 days is needed. Fever, night sweats, breathlessness, itching, or weight loss make it more likely.	NG12 2015, non-Hodgkin and Hodgkin lymphoma, pp.274 and 280
weight_loss	several cancers	talk soon	Weight loss with no known reason can be a sign of several cancers. These include bowel, stomach, gullet, lung, prostate, pancreas, kidney and bladder cancer. It can also have other causes. Guidelines say a doctor should look for the cause soon.	NG12 2015, symptoms that are not specific to one cancer, p.347
appetite_loss	several cancers	talk soon	Losing your appetite for no known reason can be a sign of several cancers. These include lung, gullet, stomach, bowel, pancreas, bladder and kidney cancer. It can also have other causes. Guidelines say a doctor should look for the cause soon.	NG12 2015, symptoms that are not specific to one cancer, p.347